Teorema Global de Peano (1890)

$J=[t_0, t_0+T]$, $f: J\times\mathbb{R}^N \longrightarrow \mathbb{R}^N$ continua y acotada

Entonces, para cada $u_0\in\mathbb{R}^N$ el P.V.I $\begin{cases} u' = f(t,u) \\ u(t_0)=u_0 \end{cases}$

tiene, al menos, una solución, $u(t;t_0,u_0)$, definida en $J=[t_0,t_0+T]$

DEM: Tonelli 1928

Esquema de Tonelli:

$$n\geq 1 \qquad u_n(t)=\begin{cases} u_0 & \text{si } t\in[t_0, t_0+\frac{T}{n}] \\ u_0+\int_{t_0}^{t-\frac{T}{n}} f(s,u_n(s))\,ds & t\geq t_0+\frac{T}{n},\ t\in J \end{cases}$$



$\mathcal{F}=\{u_n : n\geq 1\}$ Acotada y Equicontinua en $C(J;\mathbb{R}^N)$

*) $t\in[t_0, t_0+\frac{T}{n}]$, $\|u_n(t)\| = \|u_0\|$

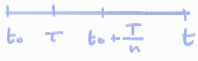
$t\geq t_0+\frac{T}{n}$, $\|u_n(t)\| = \|u_0+\int_{t_0}^{t-\frac{T}{n}} f(s,u_n(s))\,ds\| \leq \|u_0\| + \int_{t_0}^{t-\frac{T}{n}} \|f(s,u_n(s))\|\,ds \leq \|u_0\| + M\cdot(t-\frac{T}{n}-t_0) \leq \|u_0\| + MT$

$\Longrightarrow \|u_n\|_\infty \leq \|u_0\| + MT \equiv R \quad \forall n\geq 1 \Longrightarrow \mathcal{F}\subset \overline{B_R(0)}$ en $C(J;\mathbb{R}^N)$ Es acotada

*) $t,\tau\in J$ $\|u_n(t)-u_n(\tau)\| \leq L|t-\tau| \quad \forall n\geq 1$

I) 

$\|u_n(t)-u_n(\tau)\| = \|u_0-u_0\| = 0$

II) 

$\tau\in[t_0, t_0+\frac{T}{n}]$, $t\geq t_0+\frac{T}{n}$

$$\|u_n(t)-u_n(\tau)\| = \|u_0+\int_{t_0}^{t-\frac{T}{n}} f(s,u(s))ds - u_0\| \leq \int_{t_0}^{t-\frac{T}{n}} \|f(s,u_n(s))\|\,ds \leq M\cdot(t-\tfrac{T}{n}-t_0) < M(t-\tau)$$

III) 

$t,\tau \geq t_0+\frac{T}{n}$

$$\|u_n(t)-u_n(\tau)\| = \|u_0+\int_{t_0}^{t-\frac{T}{n}} f(s,u_n(s))ds - u_0 - \int_{t_0}^{\tau-\frac{T}{n}} f(s,u_n(s))\,ds\| = \|\int_{\tau-\frac{T}{n}}^{t-\frac{T}{n}} f(s,u_n(s))\,ds\| \leq M|t-\tau|$$

Son globalmente Lipschitzianas $\Longrightarrow$ son equicontinuas. $M|t-\tau|<\varepsilon$, $|t-\tau|<\delta=\frac{\varepsilon}{M}$, $\|u_n(t)-u_n(\tau)\|<\varepsilon$ si $|\tau-t|<\delta$ $\forall n\geq 1$

$\Longrightarrow$ Es acotada y equicontinua.

Por lo tanto, es relativamente compacto por el T^{ma} de Ascoli-Arzelà

Podemos extraer una subsucesión $\{u_{n_m}\}_{m\geq 1}$ uniformemente en J (convergente en $C(J;\mathbb{R}^N)$)

$$\|u_{n_m}-u\|_\infty = \max_{t\in J}\|u_{n_m}(t)-u(t)\| \longrightarrow_{n\to\infty} 0$$

$u_{n_m} \xrightarrow[n\to\infty]{} u \in C(J;\mathbb{R}^N)$ uniformemente en J

$$u_{n_m}(t) = \begin{cases} u_0 & \text{si } t \in [t_0, t - \frac{T}{n_m}] \\ u_0 + \int_{t_0}^{t-\frac{T}{n_m}} f(s, u_{n_m}(s))\,ds & \text{si } t > t_0 + \frac{T}{n_m} \end{cases} \qquad m \geq 1$$

$u_{n_m}(t_0) = u_0 \xrightarrow[n\to\infty]{} u(t_0) \qquad u(t_0) = u_0$

$t > t_0$ 

$t_0 + \frac{T}{n_m} < t$ si $m \geq m_0$

$$u_{n_m}(t) = u_0 + \int_{t_0}^{t-\frac{T}{n_m}} f(s, u_{n_m}(s))\,ds = u_0 + \int_{t_0}^{t} f(s, u_{n_m}(s))\,ds - \underbrace{\int_{t-\frac{T}{n_m}}^{t} f(s, u_{n_m}(s))\,ds}_{\to 0 \text{ si } m\to\infty} = u_{n_m}(t) \quad m \geq m_0$$

Es la definición del esquema de Euler.

$$\left\| \int_{t-\frac{T}{n_m}}^{t} f(s, u_{n_m}(s))\,ds \right\| \leq M\left(t - t + \frac{T}{n_m}\right) = M\frac{T}{n_m} \xrightarrow[m\to\infty]{} 0$$

Pasamos el lim dentro (Teorema de convergencia dominada de Lebesgue)

$$u(t) = u_0 + \lim_{m\to\infty} \int_{t_0}^{t} f(s, u_{n_m}(s))\,ds \overset{\downarrow}{=} u_0 + \int_{t_0}^{t} f(s, u(s))\,ds$$

Convergencia puntual $\lim_{m\to\infty} f(s, u_{n_m}(s)) = f(s, u(s))$

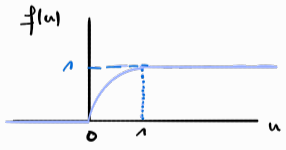
Dominación: $\| f(s, u(s)) \| \leq M$

- Hay que destacar que no hay unicidad de solución, vamos a verlo con un ejemplo:

EJEMPLO DE NO UNICIDAD EN EL MARCO DEL TEOREMA DE PEANO:

$$f(u) = \begin{cases} 0 & \text{si } u \leq 0 \\ u^q & \text{si } 0 < u \leq 1 \\ 1 & \text{si } u > 1 \end{cases} \qquad (0 < q < 1) \qquad \text{autónoma}$$

, $f'(u) = q u^{q-1} \Rightarrow f'(0) = +\infty$, $f'(1) = q < 1$



es continua y acotada

$$\begin{cases} u' = f(u) \\ u(0) = 0 \end{cases}$$ tiene al menos una solución en $[0,T]$ $\forall T > 0$

$$\begin{cases} u' = u^q \\ u(0) = \varepsilon > 0 \end{cases} \qquad u^{-q}(t)\cdot u'(t) = 1 \qquad \frac{d}{dt}\frac{u^{-q+1}(t)}{-q+1} = 1 \qquad \frac{u^{1-q}(t)}{1-q} = t + C \qquad \frac{\varepsilon^{1-q}}{1-q} = C$$

$$\frac{u^{1-q}(t)}{1-q} = t + \frac{\varepsilon^{1-q}}{1-q} \qquad u^{1-q}(t) = (1-q)t + \varepsilon^{1-q}$$

$$u(t) = [(1-q)t + \varepsilon^{1-q}]^{\frac{1}{1-q}} \quad , \quad \varphi(t) = [(1-q)t]^{\frac{1}{1-q}} \quad , \quad \varphi'(t) = \frac{1}{1-q}(1-q)[(1-q)t]^{\frac{1}{1-q}-1} = [(1-q)t]^{\frac{1-1+q}{1-q}} = \frac{q}{1+q} = \varphi^q(t)$$

$$\Rightarrow \begin{cases} u' = f(u) = u^q & \text{si } u \geq 1 \\ u(0) = 0 \end{cases} \qquad \varphi(t) = [(1-q)t]^{\frac{1}{1-q}}$$